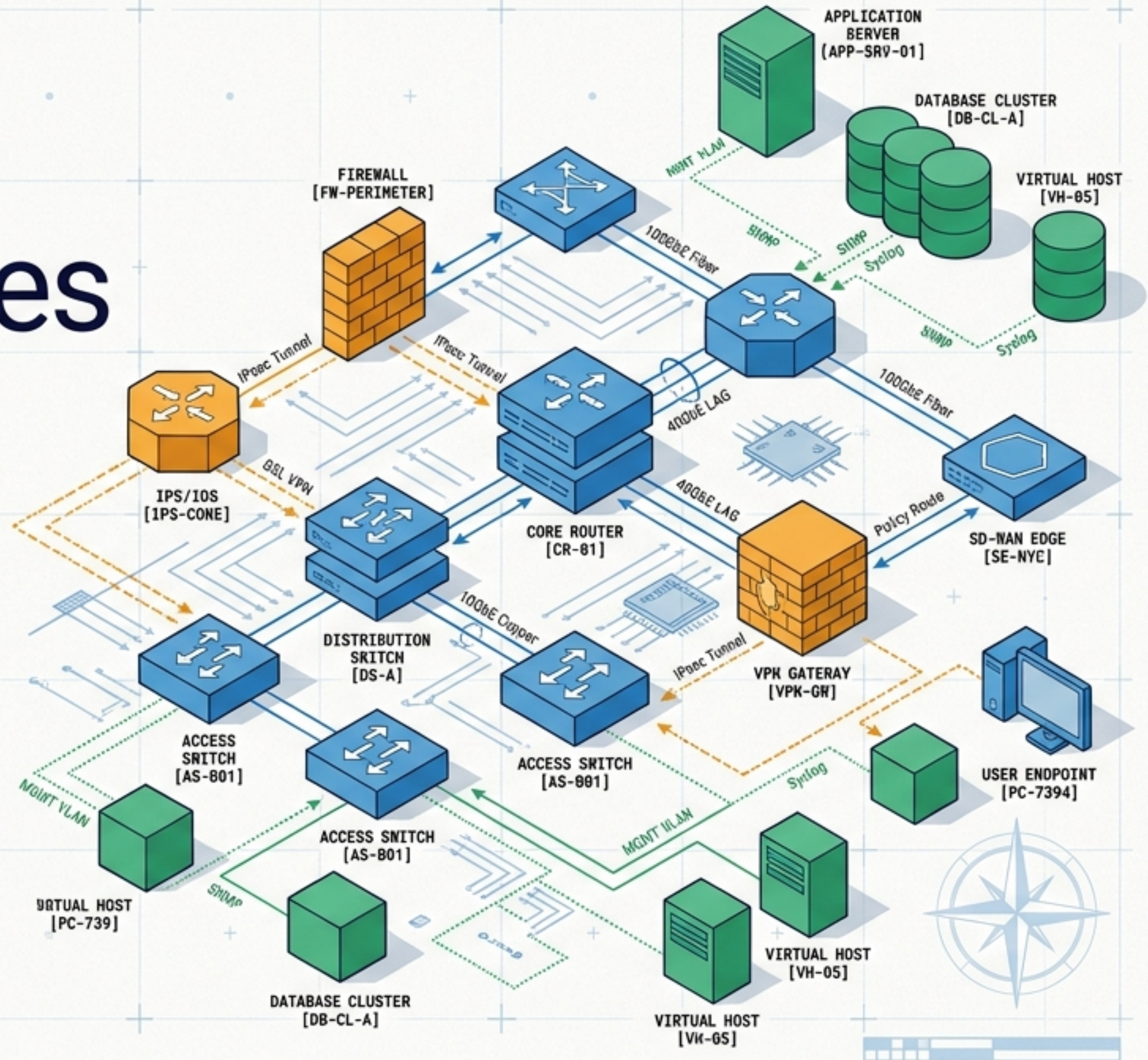


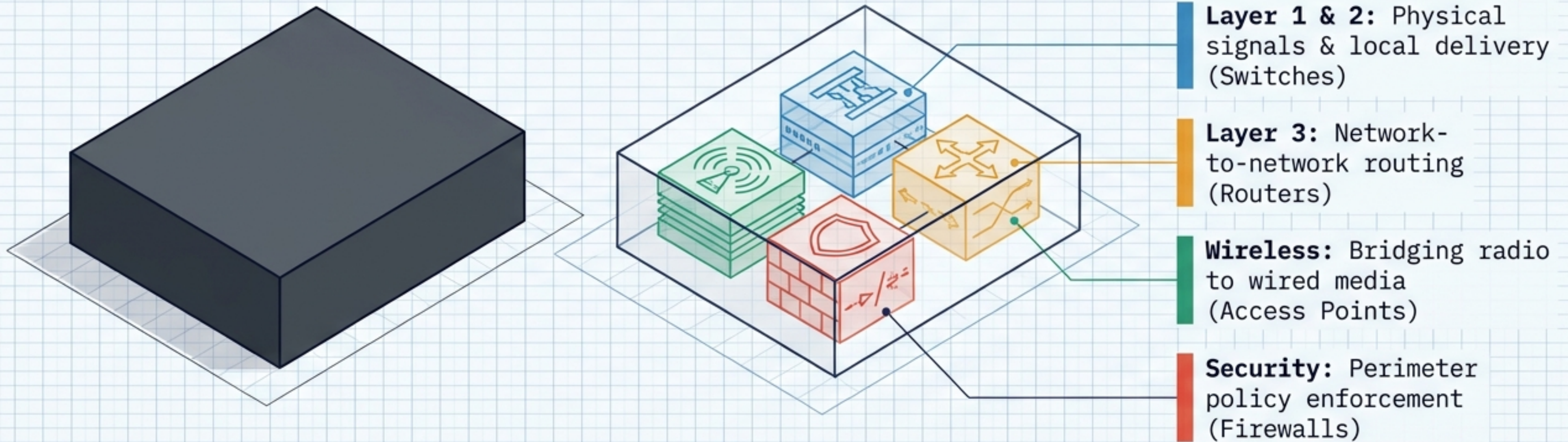
Network Devices & Functional Architecture

From physical signals to Layer 3 routing and security policy.



Form vs. Function in Network Architecture

“Do not identify a network device only by its physical box. Identify the networking functions it performs.”



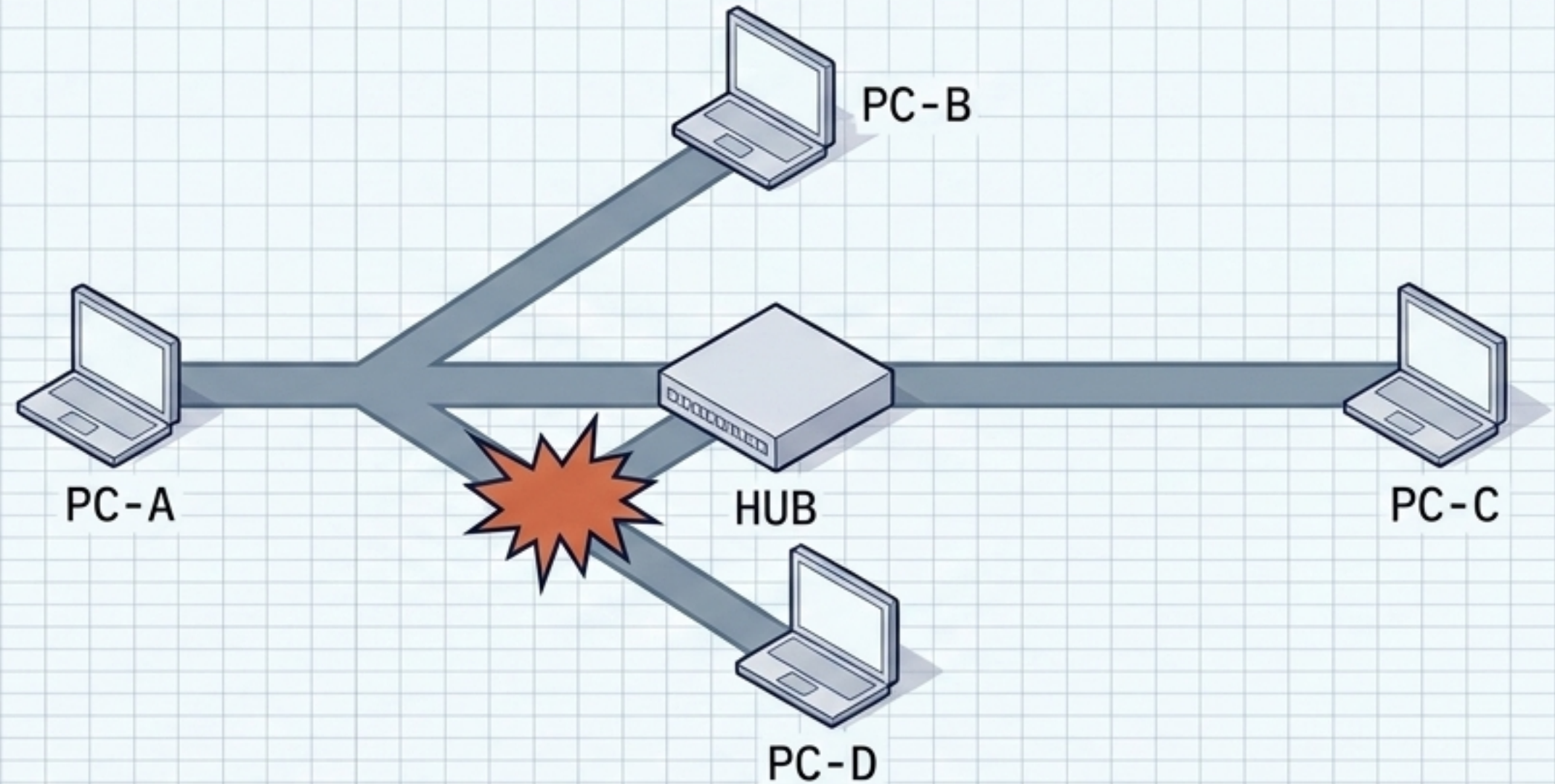
Layer 1 — The Hub (The Obsolete Relic)

Core Mechanism

Receives a physical signal

→ Repeats the signal

→ Sends to all other ports.



Key Insights

No Intelligence

A hub operates solely on physical signals. It does not understand MAC or IP addresses.

Shared Collision Domain

All connected devices share the same medium. If two transmit simultaneously, a collision occurs.

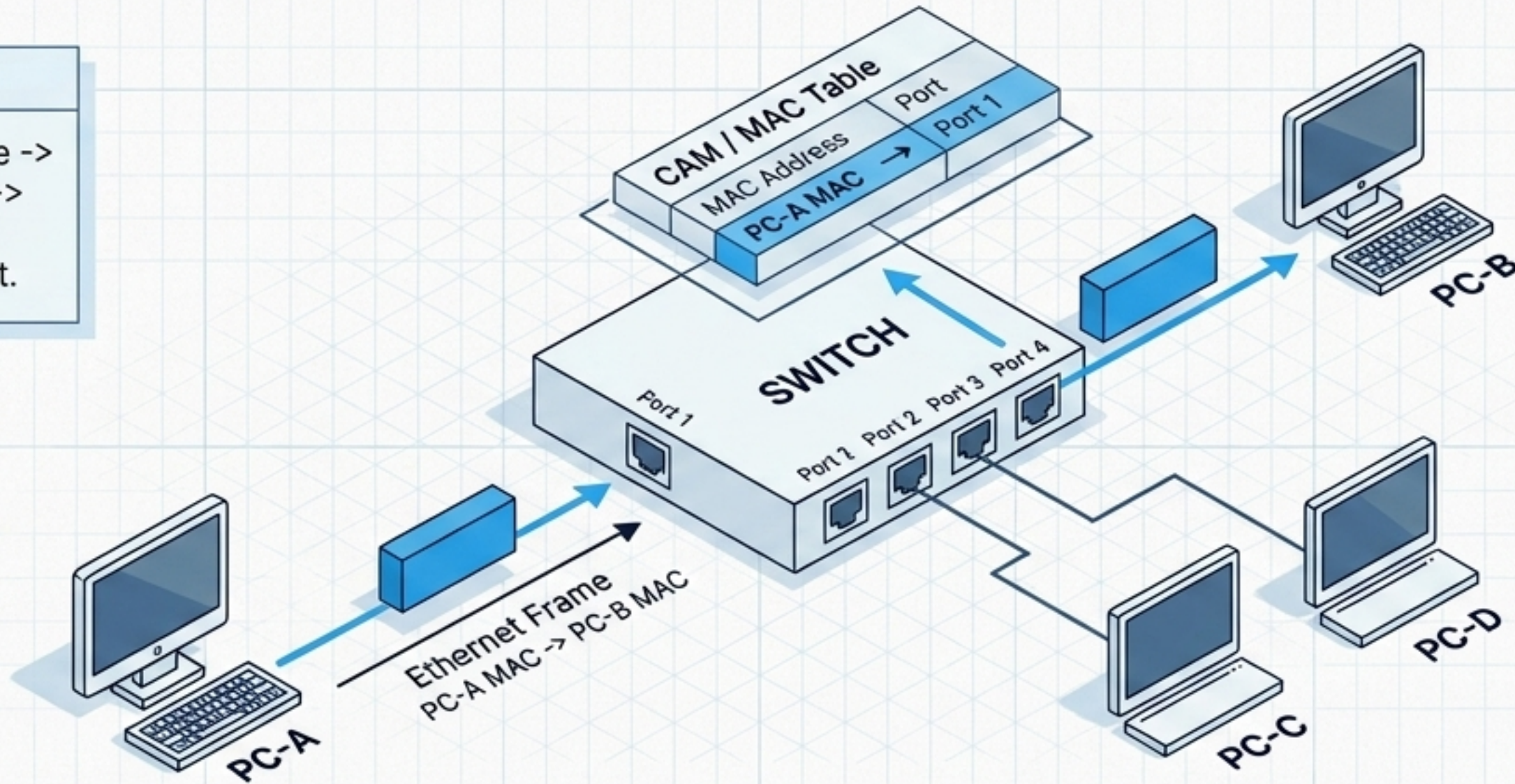
Inefficient Scaling

Every transmission reaches every device, creating unnecessary traffic and contention.

Layer 2 — The Switch (The Local Postmaster)

Core Mechanism

Receives Ethernet frame ->
Reads MAC addresses ->
Checks CAM Table ->
Forwards to correct port.



Key Insights

MAC Address Learning

The switch continuously maps Source MAC addresses to incoming ports to build its forwarding intelligence.

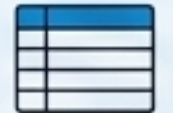
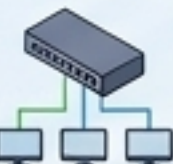
Isolated Collision Domains

Every port is a separate collision domain. Modern full-duplex links eliminate collisions entirely.

Selective Forwarding

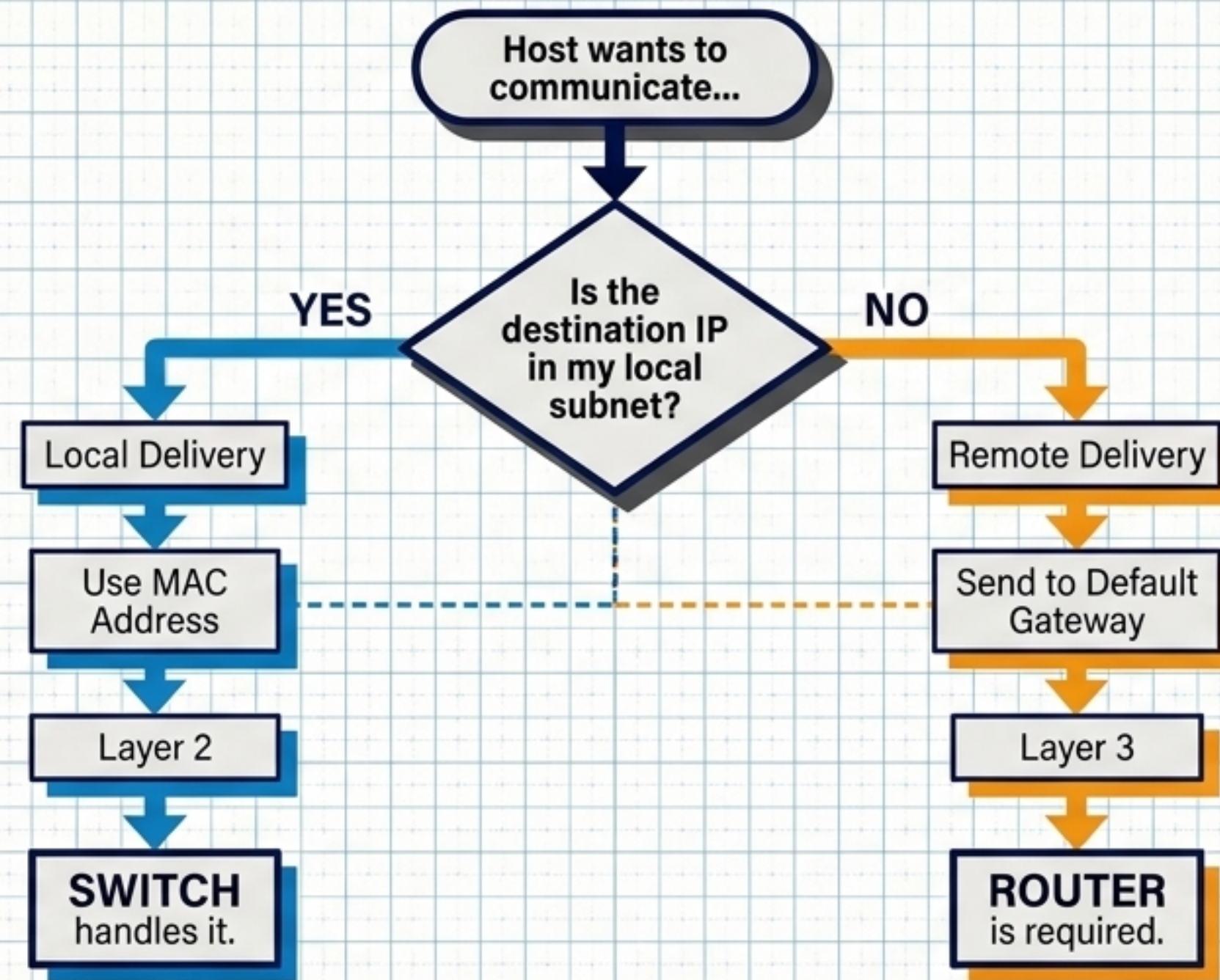
Known unicasts are sent only to the destination. (Unknown unicasts and broadcasts are flooded within the VLAN).

The Evolutionary Leap: Repeating vs. Forwarding

	Hub Slate Grey	Switch Switch
Primary Layer	Layer 1 (Physical) 	Layer 2 (Data Link) 
Main Operation	Repeats physical signals 	Forwards Ethernet frames 
Understands MAC?	✗ No	✓ Yes
Uses a Memory Table?	✗ No	✓ Yes (CAM/MAC Table) 
Collision Domain	One shared domain for all ports 	Separate domain per port 
Modern Status	Obsolete 	Standard network foundation 

Takeaway: A hub repeats; a switch forwards.

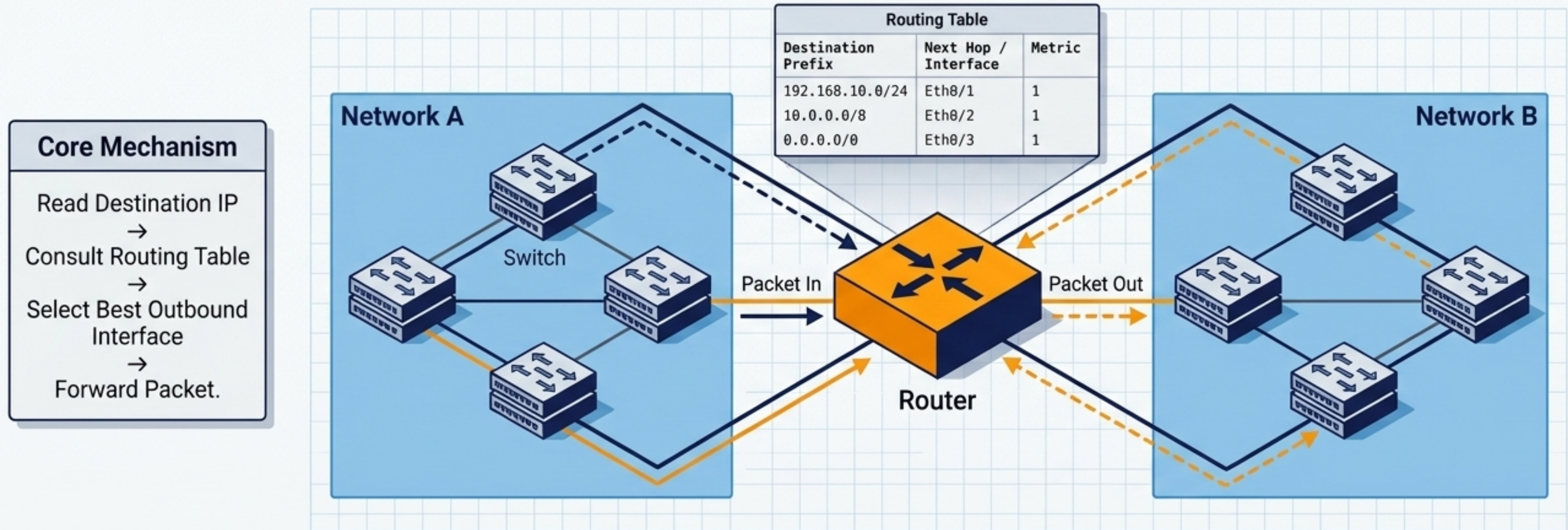
The Network Boundary: The Same Subnet Test



Key Insight:

This single logical test dictates whether a network will utilize MAC-based local delivery or IP-based routing.

Layer 3 — The Router (The Network Navigator)



Key Insights

Network-to-Network

Routers connect different IP networks (subnets), not just individual devices.

Longest Prefix Match

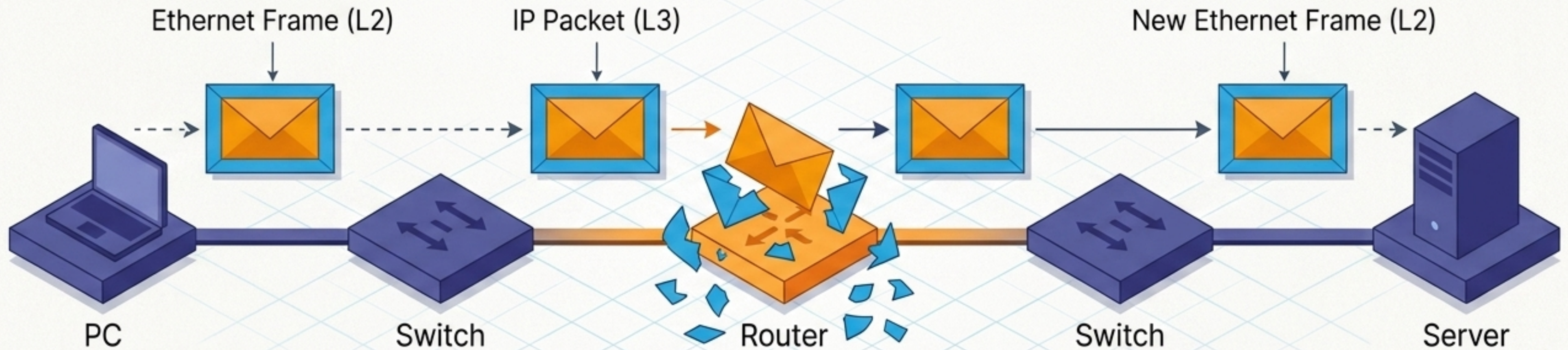
When multiple routes match a destination, the router strictly follows the most specific matching route (e.g., prefers /24 over /8).

Separates Broadcast Domains

Layer 2 broadcast traffic is contained within the local switch network and is not forwarded across router interfaces.

Packet Walk: End-to-End IP, Hop-to-Hop MAC

Concept: MAC addresses and IP addresses solve different problems. They do not replace one another.



The IP Packet (Layer 3)

The IP Packet (Layer 3) is the letter. It travels end-to-end (Source PC to Destination Server).

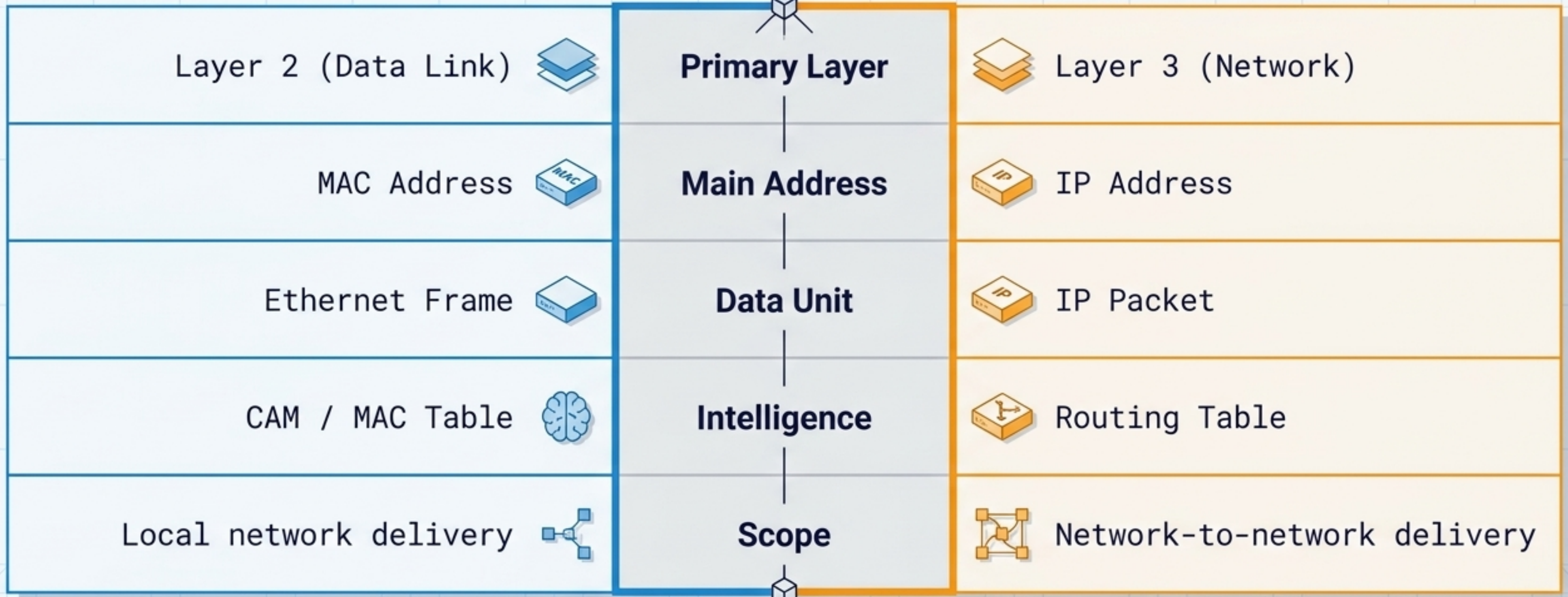
The Ethernet Frame (Layer 2)

The Ethernet Frame (Layer 2) is the delivery envelope. It is strictly local. The router opens the frame, reads the IP packet, and creates a completely new Ethernet frame for the next hop.

The Definitive Guide: Switch vs. Router

Switch asks: "Which local port holds this MAC?"

Router asks: "Which path leads to this IP network?"



The Access Point (Extending Layer 2)

Core Mechanism

Receives wireless signal
→ Bridges radio to wired medium
→ Forwards to Ethernet switch.



Wireless Bridging, Not Routing

A dedicated AP simply connects wireless clients into the existing wired Layer 2 network.

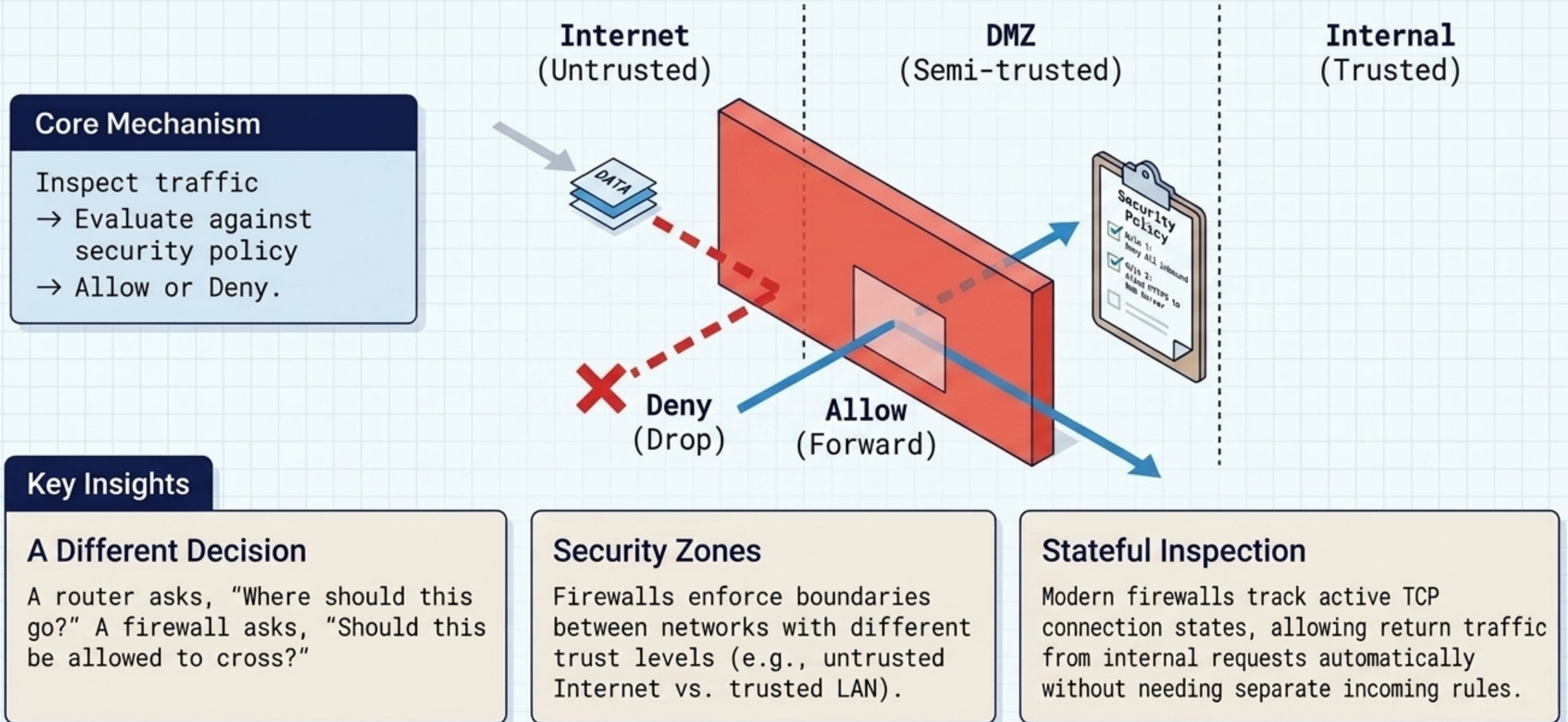
Shared Radio Medium

Unlike isolated full-duplex switch ports, wireless operates in a shared radio environment requiring unique access control.

MAC Addresses Apply

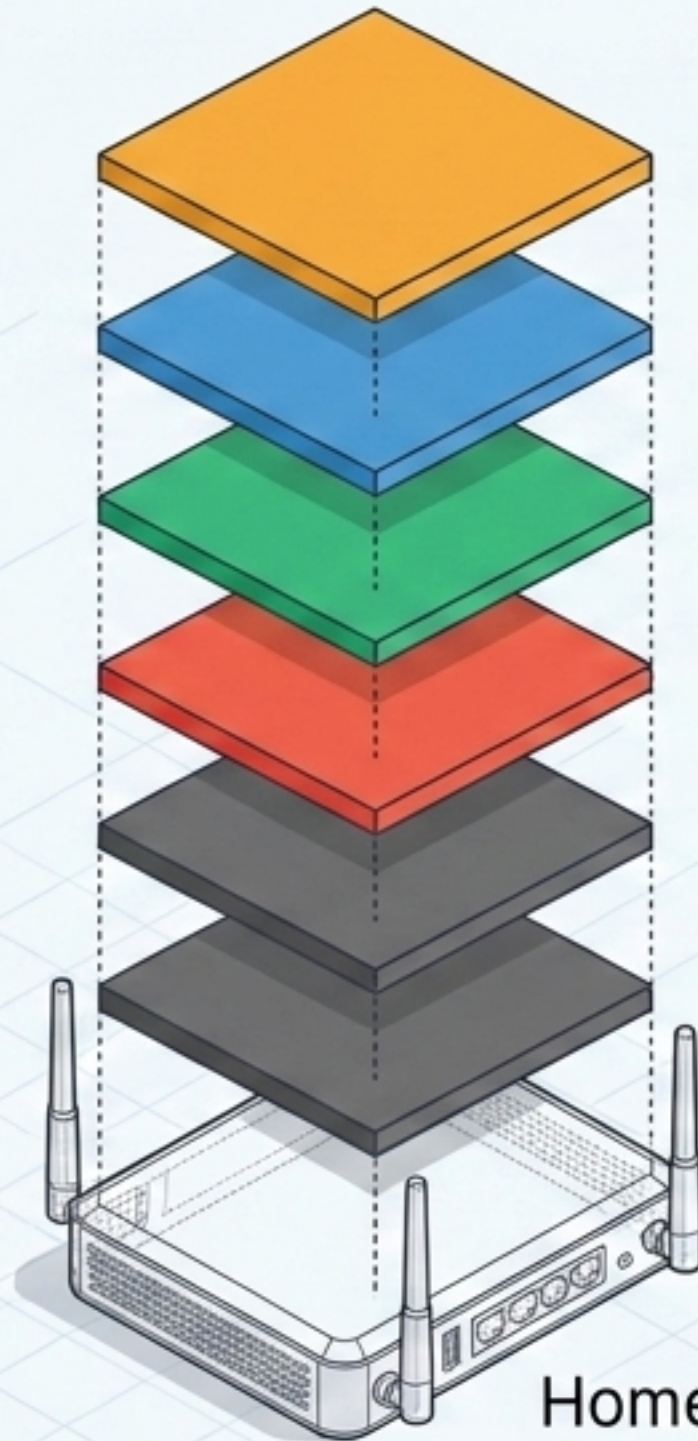
Wireless devices still possess MAC addresses and participate natively in Layer 2 communication.

The Firewall (The Security Gatekeeper)



Synthesis: Anatomy of a Home Gateway

Insight: Consumer devices collapse multiple distinct enterprise functions into a single physical chassis.



1. **Router** - Connects the home subnet to the ISP network.
2. **Ethernet Switch** - Provides local connectivity for the physical LAN ports.
3. **Access Point** - Bridges Wi-Fi clients to the LAN.
4. **Firewall** - Protects the home network from uninitiated Internet traffic.
5. **NAT** - Translates internal private IPs into a single public IP.
6. **DHCP Server** - Automatically assigns IP configurations to joining devices.

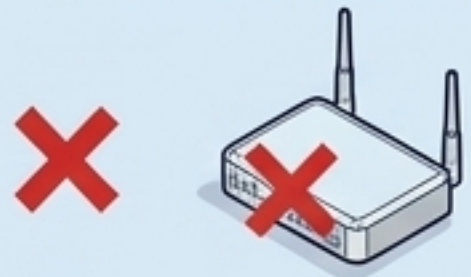
Home Wi-Fi Box

Troubleshooting by Functional Role

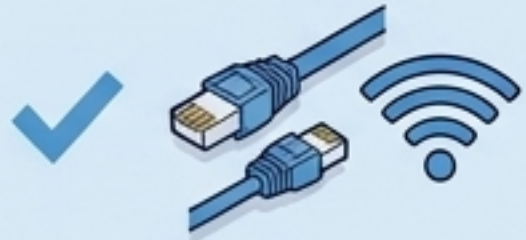
Concept: Do not randomly blame “the network.” Systematically isolate the failure layer based on the destination.

Same-Subnet Failure

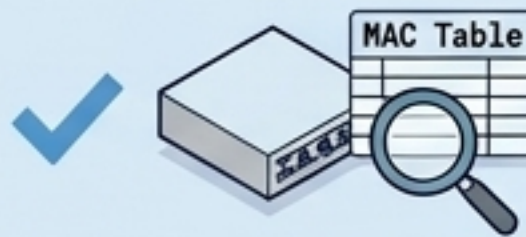
(e.g., 192.168.1.10 to 192.168.1.20)



Do NOT check the Internet router.



Check: Physical cables / Wi-Fi connection.



Check: Local Switch CAM table / MAC learning.



Check: Host-level software firewalls.

Different-Subnet Failure

(e.g., 192.168.1.10 to 8.8.8.8)



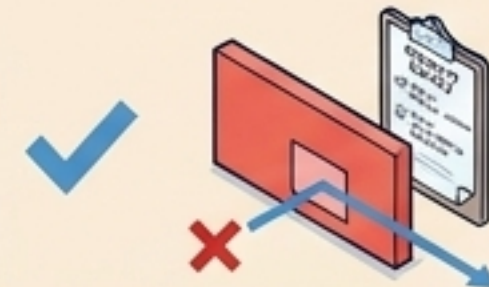
Check: Is the Default Gateway reachable?



Check: Does the Router's Routing Table have a path?

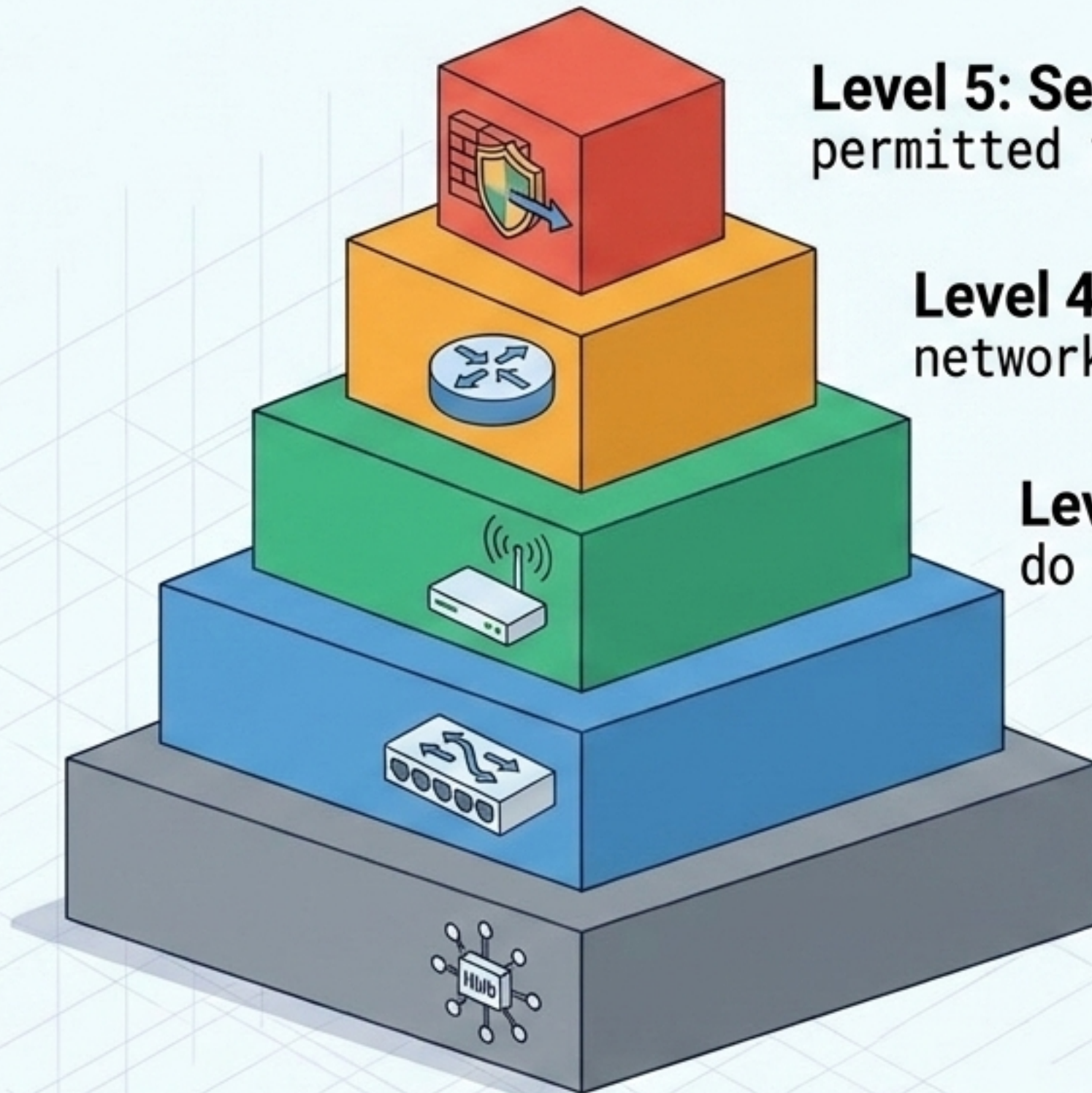


Check: Is there a valid return route?



Check: Is a perimeter Firewall policy blocking the protocol?

The Layered Device Framework



Level 5: Security (Firewall) -> "Is this traffic permitted to cross the boundary?"

Level 4: Network (Router) -> "Which IP network should this packet go to?"

Level 3: Wireless (Access Point) -> "How do wireless devices join the LAN?"

Level 2: Data Link (Switch) -> "Which local port gets this MAC address?"

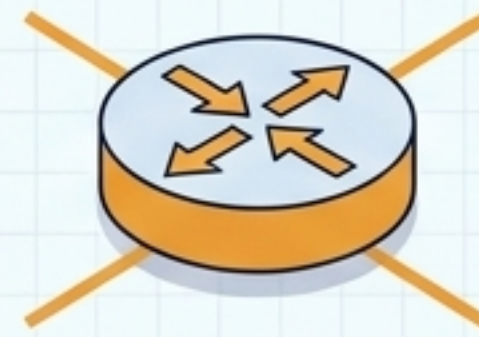
Level 1: Physical (Hub) -> "How do I repeat this physical signal?"

Final Mental Model



SWITCH

- Layer 2
- MAC Addresses
- Local Forwarding



ROUTER

- Layer 3
- IP Addresses
- Network Routing



ACCESS POINT

- Wireless Layer 2
- Wi-Fi
- LAN Bridging



FIREWALL

- Security Boundary
- Policy Rules
- Traffic Inspection

The Golden Rules:

- Same subnet = Switch / MAC delivery.
- Different subnet = Router / IP delivery.
- Physical box does not equal logical function.

Master the functions, and the physical network will always make sense.